

Design Verification Plan & Report — VBF-T6R2-DVPR-001

Design Verification Plan & Report (VBF-T6R2-DVPR-001)

****Case**:** exp/t6_r2 · ****Candidate**:** R12a_h140 · ****Date**:** 2026-08-26
Verification performed in-simulation per the Virtual Battery Factory protocol; each item maps to a pre-registered criterion in log.jsonl entry 0 (not post-hoc revisable).

#	Requirement (task text)	Pre-registered threshold
1	Volumetric energy density ≥ 950 Wh/L	energy_density_wh_l ≥ 950
2	Voltage plateau ≥ 4.1 V	midpoint_voltage_v ≥ 4.1
3	Max temperature ≤ 50 °C	T_max_K ≤ 323.15
4	4C fast charge, no lithium plating	plated == False (any anode_p)
5	Anode SEI ≤ 500 nm after 100 cycles	sei_thickness_nm_end ≤ 500

Evidence files: cell/r12_a_energy_dfn.json, cell/r12_a_1c_dfn.json, cell/r12_a_4c45_dfn.json, cell/r12_a_aging.json. Mechanical verdict recorded by `bda log-evaluate` round 12 (candidate R12a_full_verification_DFN, verdict=pass, 5/5 checked). Stage-1 molecular criteria (max_energy_ev, max_homo_ev) are not in scope for a start_stage=3 case (unchecked).

Test condition notes (honest disclosure)

- The 4C protocol current is $4 \times$ system-file nominal capacity (4.5 Ah) = 18 A; actual simulated 1C capacity is 6.374 Ah $\square$ the executed charge is ~2.8C effective. A follow-up verification at true 4C (25.5 A) is recommended and listed in DFMEA as an open risk.
- Aging protocol per-cycle capacity outputs are negative artifacts of the LNMO file parameterization; the SEI criterion is judged on sei_thickness_nm_end as pre-registered.
- T_max margin (0.45 K) is the binding constraint; the thermal-interface spec ($A \geq 0.0106 \text{ m}^2$, $h \geq 140 \text{ W/m}^2\cdot\text{K}$) is a hard requirement, not a nominal suggestion.
- SPMc cross-check results (ED 1106.3, midV 4.2202, T 320.30 K, not plated) are consistent; the DFN values above are the acceptance basis.

Verification environment

- Engine: PyBaMM (Chen2020 + LNMO system file), SPMc proxy and DFN precise models.
- True compute (DFT/MD) skipped per pre-registered real_compute=false — endorsement record in log.jsonl states this explicitly.